

Quarto Manuals

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Abstract

to be fleshed out from bullet form below

Impact Statement

Scientific programming workflows are becoming increasingly complex, while existing support mechanisms remain poorly matched to that complexity. Traditional documentation is flexible but non-executable, command-line wrappers are executable but often too rigid, and current agentic systems are promising but not yet sufficiently reliable for many high-stakes workflow setup tasks. We propose Quarto Manuals, an executable documentation framework built on Quarto that helps authors create flexible, scalable, and reproducible workflow manuals composed of structured pages with explicit procedures and checks. Powered by the quarto-emit backend, Quarto Manuals enable authors to create executable documentation that allows users to conveniently step through scientific computing workflows while generating reproducible project artifacts and receiving immediate feedback on whether each step has been implemented correctly.

Keywords: APA Style, Quarto, Markdown, apaquarto

Quarto Manuals

Abstract

- Programming is already difficult, and research data workflows are growing in scale and complexity [CITE NEEDED]. Scientists need workflow rigor, not just working code.
- Traditional documentation helps, but it breaks down for large, multi-step, order-dependent workflows. It is not executable, reproducible, or able to confirm that a user is proceeding correctly. [CITE PROBABLY NEEDED]
- Command-line wrappers are executable, but often too rigid: either the workflow is over-constrained, or flexibility is pushed into an unwieldy number of flags and options.
- AI and agentic systems are promising, but today they are still limited by (FACT) non-determinism [CITE NEEDED] and (OPINION) uneven reliability in complex scientific workflows (though we are working to address these issues with agentic workflows).
- We propose **Quarto Manuals**, a framework for executable software manuals built on Quarto starter templates and extensions.
- A Quarto Manual guides users through a workflow as ordered, interactive pages that combine explanation, code execution, and verification.
- Manuals are flexible across audiences, projects, organizations, and programming languages, while remaining reproducible because their side effects are explicit and inspectable.
- Rather than hiding workflow decisions, Quarto Manuals keep them visible and editable, which is especially important in research computing environments with local constraints and judgment calls.
- The backend is powered by `quarto-emit`, a lightweight extension that materializes conventional workflow artifacts from manual pages when needed.
- We demonstrate the approach with three examples of increasing sophistication: a simple manual, an intermediate manual, and a complex manual.
- For authors, Quarto Manuals provide a structured but adaptable template for turning complex recurring workflows into reusable manuals.

- For operators, they provide a stepwise, testable process for leaving behind reproducible setup artifacts and confirming progress in a structured way through the workflow.

Feature Roadmap

- ☐ abstract as prose
- ☐ stylize each block so it is visually distinct
- ☐ use Lua to print out useful render messages when blocks are being processed
- ☐ can Lua be used to connect checks and prereqs across pages? Maybe?
- ☐ setup a testing suite in a separate repo

Hypotheses, Aims, and Objectives

The last paragraph of the introduction usually states the specific hypotheses of the study, often in a way that links them to the research design.

Method

General remarks on method. This paragraph is optional. Not all papers require each of these sections. Edit them as needed. Consult the [Journal Article Reporting Standards](#) for what is needed for your type of article.

Participants

Who are they? How were they recruited? Report criteria for participant inclusion and exclusion. Perhaps some basic demographic stats are in order. A table is a great way to avoid repetition in statistical reporting.

Measures

This section can also be titled **Materials** or **Apparatus**. Whatever tools, equipment, or measurement devices used in the study should be described.

Measure A

Describe Measure A.

Measure B

Describe Measure B.

Subscale B1. A paragraph after a 4th-level header will appear on the same line as the header.

Subscale B2. A paragraph after a 4th-level header will appear on the same line as the header.

Subscale B2a. A paragraph after a 5th-level header will appear on the same line as the header.

Subscale B2b. A paragraph after a 5th-level header will appear on the same line as the header.

Procedure

What did participants do? How are the data going to be analyzed?

Results**Descriptive Statistics**

Describe the basic characteristics of the primary variables. My ideal is to describe the variables well enough that someone conducting a meta-analysis can include the study without needing to ask for additional information.

Table 1 is an example of a plain markdown table. Note the that the caption begins with a colon.

Table 1

My Caption.

Letters	Numbers
A	1
B	2
C	3

Note. My note

Discussion

Describe results in non-statistical terms.

Limitations and Future Directions

Every study has limitations. Based on this study, some additional steps might include...

Conclusion

Describe the main point of the paper.

References

References marked with an asterisk indicate studies included in the meta-analysis.